Analysing the temperature trends for coastal and non-coastal measuring stations in the 40-50 degree latitude band

1. Introduction

Many climate models predict that land warms faster than ocean, and will continue to do so with increasing global temperatures. The large heat capacity of water, the heat transportation down the water column, and water evaporation are all factors contributing to this event. One can thus assume that coastal regions will warm less than non-coastal regions with similar climates and at similar latitudes. This analysis tests this by comparing the temperature data of several measuring stations in Europe and West-Asia between 1950 and 2016.

2. Methodology

The initial dataset was fetched from the climate time series browser (1), and consisted of all the measuring stations between 40 and 50 degree latitude, -10 and 90 degree longitude, and with a sea level below 500m. The latitude has to be bounded, since temperature drops significantly with higher latitudes. Higher altitude measuring stations were removed due to the temperature dropping ~6.5 deg C per 1000m altitude, following the atmospheric lapse rate. The temperature data was normalised by subtracting the average value between 1900 and 1950 of each record.

The data was further processed in Python to remove a. all measuring stations without data and b. measuring stations with less than 50% of data points between 1950 and 2016. Since the downloaded data did not provide coordinates for the measuring stations, the coordinates were geocoded (get coordinates from location name) from the location names using the open street map API. Additional checks were implemented to make sure the coordinates were bound to latitudes and longitudes of interest, and to only include names of cities and places (no companies or similar).

Lastly, from the estimated measuring station coordinates, the distance to the nearest coastline was calculated using Natural Earth's 10m coastline map (2). This map includes the Caspian Sea, even though it is technically a lake. Measuring stations were binned in 3 categories: coastal (<50km to nearest coast), near-coastal (between 50 and 500km) and inland (>500km). Additionally, the temperature trends between 1950 and 2016 per measuring station were calculated.

3. Results

Out of the initial 353 measuring stations fetched (1), several had to be removed. 247 were removed due to only containing NaN values, 22 were removed because less than 50% data points were present in the time series between 1950 and 2016 to calculate the trend, and 1 had to be removed because open street map was unable to find the coordinates for that measuring stations. This resulted in 83 measuring stations analysed after preprocessing the dataset.

Figure 1 provides a map of the station locations, with their respective binned coastal type and the temperature trend in deg C/decade. Low and negative temperature trends between -0.1 and 0.1 deg

C/decade, can be found around the black sea and Mediterranean. Higher temperature trends of up to 0.4 deg C/decade can be found further inland, in particular in Kazakhstan.

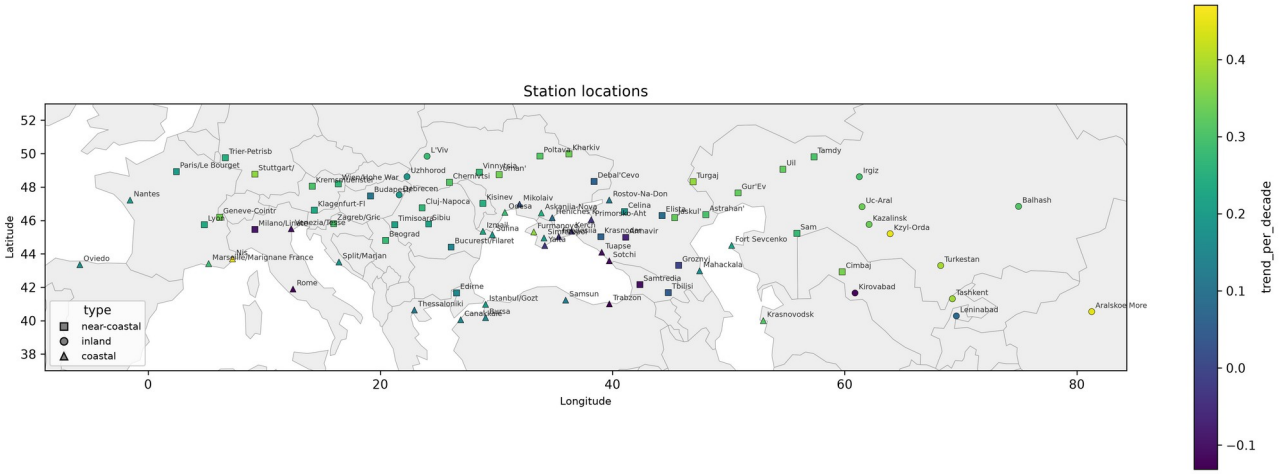


Figure 1: Overview of the measuring station locations, their classified coatal type and the temperature trend per decade (in deg C / decade)

One can refer to Table 1 for an overview of the statistical quantities per type. With 13 records for the inland type, it is slightly underrepresented compared to the other types. A clear visible increase in the mean temperature trend is visible per type, however. Starting from 0.12degC/decade for coastal locations, increasing to 0.22 for near-coastal and finishing with 0.27 for inland locations. The trend difference between coastal and near-coastal is much higher than the one from near-coastal to inland. This can be attributed to the fact that the influence of bodies of water reduces with distance, and plateaus from a certain distance. This was also confirmed by a Tukey HSD test, showing that coastal stations warmed significantly less than near-coastal (+0.10 degC/decade, p = 0.02) and inland stations (+0.15 degC/decade, p = 0.005), while near-coastal and inland stations did not differ significantly (p = 0.43).

Table 1: Statistical descriptors for the temperature trend per decade and the coastal distances, grouped by type

		Trend/decade [deg C/decade]				**Coastal distance [km]**			
	Count	**Mean**	**Std**	**Min**	**Max**	**Mean**	**Std**	**Min**	**Max**
Type									
coastal	31	0.12	0.16	-0.13	0.47	10.97	14.09	0.06	39.57
near-coastal	39	0.22	0.13	-0.11	0.38	249.99	129.47	52.09	486.34
inland	13	0.27	0.16	-0.12	0.45	1016.93	547.57	517	2166.11

Lastly, Figure 2 presents the temperature trend in function of the distance to coast, grouped by type. There is quite a wide spread on the temperature trends in the coastal (std=0.16degC/decade) and first half of near-coastal (std=0.13degC/decade) locations. Many urban environments reside in these areas. It is a well known fact that urban environments have many artificial heating and cooling effects, such as urban heat island effect (heating), heating from residential and industry (heating) and greater aerosol concentrations in the atmosphere in these regions (cooling). Moreover, since the coordinates for the measuring stations were geocoded instead of using the exact coordinates, the coastal distances are an estimation and might skew the results.

Nevertheless, a clear upward trend can be seen of 0.15degC/decade/1000km with a 95% confidence interval of [0.07, 0.23], showing that coastal regions indeed warm less than regions further away from the coast.

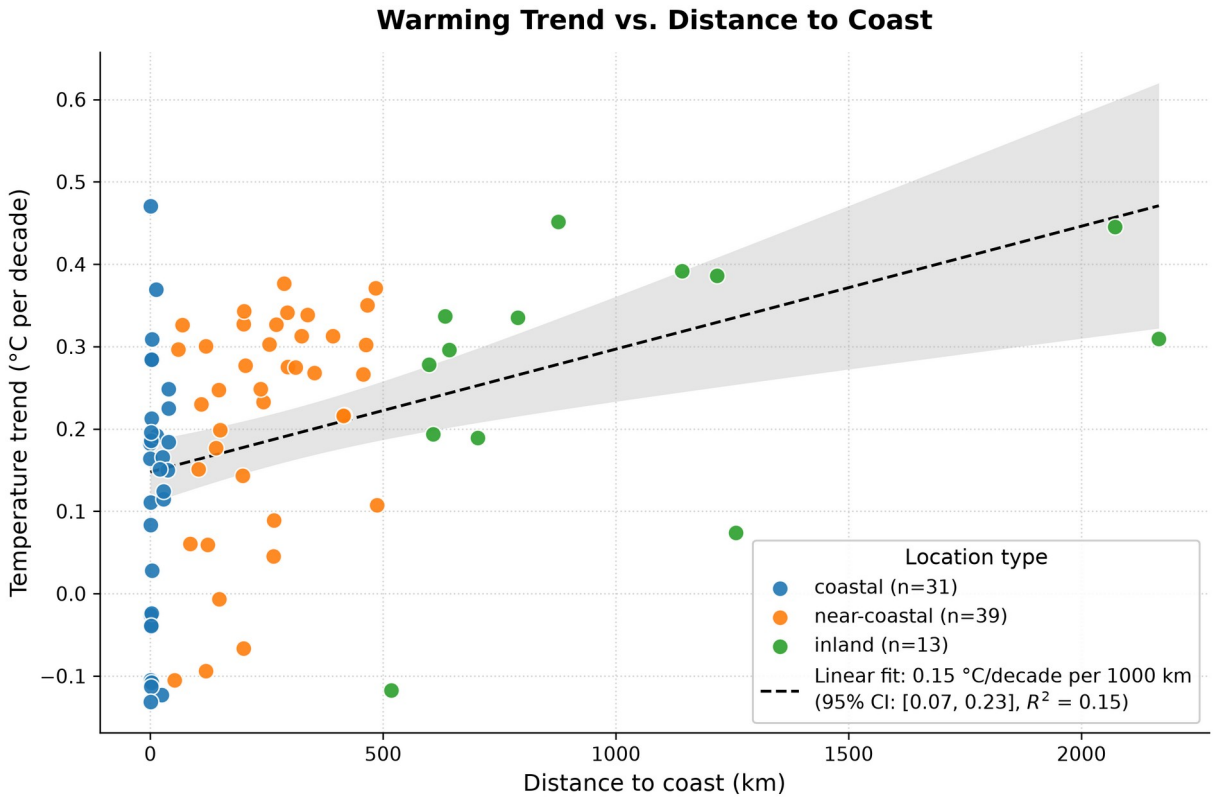


Figure 2: Scatter plot of temperature trends vs coastal distance for every measuring location, grouped by type

4. Conclusions

From the results it can be concluded that coastal regions have warmed less over 1950-2016 than non-coastal regions with similar climates and at similar latitudes. The temperature data of 83 measuring stations was analysed, and the measuring stations binned in 3 types according to their distance to the coast: a) coastal (<50km), b) near-coastal (between 50 and 500km), and c) inland (>500km). Results show that the mean temperature trend per decade per type indeed increases. Furthermore, a temperature trend of 0.15 degC/decade/1000km was found. Both consistent with the hypothesis.

This study did not take into account any artificial heating effects in urban environments. Future works should take these into account and decouple them from natural heating effects to get a full understanding of the impact. Moreover, the bias introduced in the selection of the measuring stations should be removed. Firstly, the exact measuring station locations should be used if this data is available. Secondly, the inland stations are mostly in Central-Asia, where regional warming might also play an important role. Thirdly, 247 out of the 353 measuring stations had no usable data, skewing the results toward long-record stations. And lastly, the Caspian Sea was included here even though it is a lake. Lastly, these trends should be compared to existing AR5 climate model outputs.

5. Sources

(1)
https://climatemodels.uchicago.edu/timeseries/#BbbBBECBBBBBBDBBBBBGCGDeBBBBDBB
EBBBBBDBBBBBBBEBBBBBKbBCBCCBBCCCEwBBBBBBBBBBBBBBBBBBBBEBBBB
FpCBBBBBBBBBBFDBBNqBBBBBBBBCBBBBmBlJBBUyBBBDTBCBJpBBBBBBBBBBBBD
veBgiBBEBCBDBCCBBGBCCEBBBBECBBBBCBBBBBBBQCBjBBBBBBBBBBBBBBBBBC
BBBBBDCEBBBBBBBBBBCcBBBBCDCBBBBBBBBBCBBDBOBBBBBBBBhBCBCCBBBB
BBBBBBBBCBBBBCBBCBBBBBBBBBBBBBBBBDBBCBBBBCBCBBBBVBBBBOMBBCC
BBDnCCBKBBBGECBBBBCBBDBBBsBCBBBBBBBBBBBBBBCBBBBBBBBBBBBBCCBB
BBBBBFCBBBBBBBBFBCBqCDBELBBDBBBBBBBBBDBBCBBBDBBBBBBBBBBBBBB
BCBDBFZBDBBBBBBGnBBBBBCBBBBBBBBBBBBBBBCf

(2)
https://www.naturalearthdata.com/downloads/10m-physical-vectors/10m-coastline/